

DONGWON JUNG

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Education

University of California Davis

June 2029

Ph.D in Computer Science

University of Southern California

May 2024

Master of Computer Science

GPA: 4.0/4.0

Sungkyunkwan University

Aug 2022

Bachelor of Computer Science and Engineering

GPA: 3.7/4.0

Experience

Microsoft Research

Jun 2026 – Sep 2026

Research Intern

Seattle, WA

- Developed a self-improving agent framework that converts its own failure trajectory into a dense supervision signal.

Amazon AWS

Jun 2025 – Sep 2025

Applied Scientist Intern

New York, NY

- Developed a novel framework that optimizes test-time computation in multi-agent systems under budget constraints.
- Proposed a planning algorithm that enables agents to collaborate effectively while allocating resources adaptively.
- Achieved up to 12% higher accuracy and significantly improved budget efficiency across multi-agent benchmarks.

Wrtn Technologies

May 2023 – Aug 2023

Machine Learning Engineer

Seoul, South Korea

- Built **LLM**-powered planner agent that plans and executes external plugin services corresponding to user's ask.
- Designed and implemented quantitative evaluation metrics to measure performance of GPT-powered planner agent.
- Constructed data augmentation flywheel that boosts the performance of the planner agent up to **13%** accuracy.

Sungkyunkwan University

Dec 2020 – Aug 2022

Undergraduate Research Intern

Suwon, South Korea

- Devised a knowledge-based recommendation model using Pytorch to alleviate cold start problem in the real world.
- Published a conference paper in Korean Institute of Intelligent Systems and won the Undergraduate Thesis Award.

Publications

- **Dongwon Jung**, Peng Shi, Yi Zhang, **FutureWeaver: Planning Test-Time Compute for Multi-Agent Systems with Modularized Collaboration**. CoLM 2026.
- Wenjie Jacky Mo, Qin Liu, Xiaofei Wen, **Dongwon Jung**, Hadi Askari, Wenxuan Zhou, Zhe Zhao, Muhao Chen, **RedCoder: Automated Multi-Turn Red Teaming for Code LLMs**. ACL, 2026.
- Jiyeon Pyo, Yuankun Jiao, **Dongwon Jung**, Zekun Li, Leeje Jang, Sofia Kirsanova, Jina Kim, Yijun Lin, Qin Liu, Junyi Xie, Hadi Askari, Nan Xu, Muhao Chen, Yao-Yi Chiang, **FRIEDA: Benchmarking Multi-Step Cartographic Reasoning in Vision-Language Models**. ICLR, 2026.
- **Dongwon Jung**, Wenxuan Zhou, Muhao Chen, **Code Execution as Grounded Supervision for LLM Reasoning**. EMNLP, 2025.
- **Dongwon Jung**, Qin Liu, Tenghao Huang, Ben Zhou, Muhao Chen, **Familiarity-aware Evidence Compression for Retrieval Augmented Generation**. EMNLP - Findings, 2025.
- Tenghao Huang, **Dongwon Jung**, Muhao Chen, **Planning and Editing What You Retrieve for Enhanced Tool Learning**. NAACL - Findings, 2024.

Projects

Transportation Agent for CalTrans

Dec 2025 – Present

- Build a LLM-based Agent System that writes grant proposals for local transportation agencies seeking federal funding.
- Construct an agent pipeline for automating CalTrans Transportation Management Plan generation.

Awards/Scholarships

Awards: Undergraduate Thesis Award (2022), Ministry of Science and ICT Award (2020), President of Korea Intelligent IoT Association Award (2020)

Scholarships: Undergraduate Full Scholarship (2016 – 2022), SKKU Dean's List (2017, 2021)